

HISAR SCHOOL

AP CSP 2025-2026

PELIN BAŞYURT

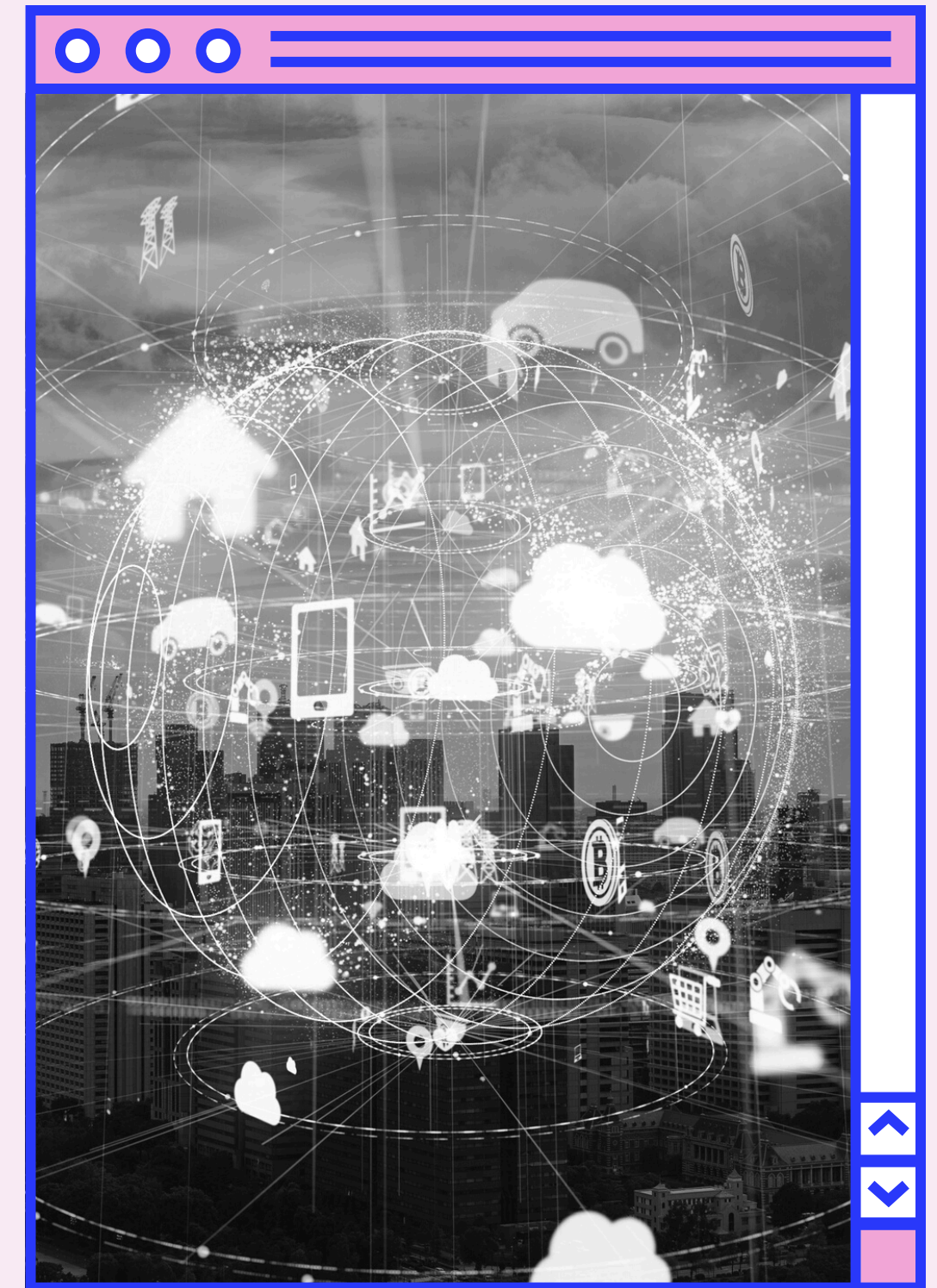
AP CSP BIG IDEA 4.1: THE INTERNET

How computers communicate across the world through
interconnected networks and protocols

BIG IDEA 4.1

THE INTERNET

The Internet is one of the most important computing systems in the world. It allows people and devices to communicate, share information, and access services across long distances. To understand how modern technology works, you need to understand how the Internet works.



AP CSP BIG IDEA 4.1



WHAT IS THE INTERNET & WHY IT MATTERS

The Internet is a network of networks. It connects many smaller networks, such as home networks, school networks, business networks, and data centers. Together, these networks allow billions of devices to communicate.

People use the Internet for messaging, video calls, streaming, gaming, research, shopping, and learning. Businesses, schools, and governments also rely on it every day. The Internet makes communication and access to information fast and widespread.

NETWORKS AND DEVICES

A NETWORK IS A GROUP OF CONNECTED COMPUTING DEVICES. THESE DEVICES CAN INCLUDE PHONES, COMPUTERS, TABLETS, SERVERS, AND PRINTERS. SOME DEVICES MAINLY SEND AND RECEIVE DATA, WHILE OTHERS HELP MOVE DATA THROUGH THE NETWORK.

EVERY DEVICE CONNECTED TO A NETWORK PLAYS A ROLE. SOME DEVICES ARE USED BY PEOPLE DIRECTLY, LIKE PHONES AND LAPTOPS. OTHERS, LIKE SERVERS, STORE AND PROVIDE INFORMATION. DEVICES SUCH AS ROUTERS HELP MOVE DATA THROUGH THE NETWORK SO COMMUNICATION CAN HAPPEN.

PACKETS AND PACKET SWITCHING

Data sent over the Internet is broken into small pieces called packets. This is called packet switching. Each packet carries part of the data, along with information about where it came from and where it is going.

At the destination, the packets are put back together in the correct order. Packet switching helps the Internet work efficiently because many users can share the network at the same time, and packets can be rerouted if one path is busy or unavailable.

AP CSP BIG IDEA 4.1



Routers, paths, redundancy, and fault tolerance

AP CSP BIG IDEA 4.1



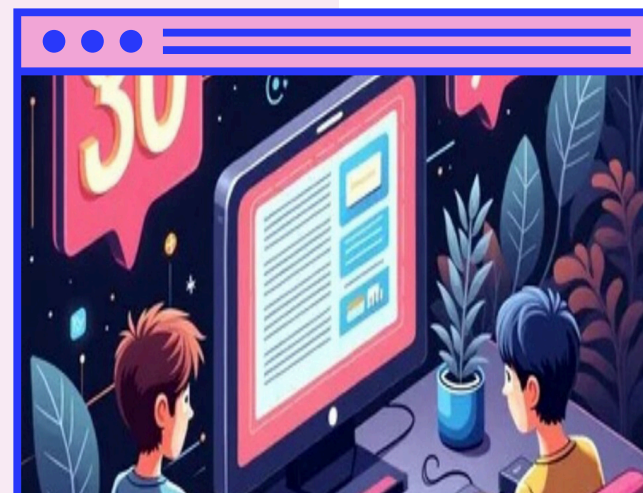
ROUTERS MOVE PACKETS FROM ONE NETWORK TO ANOTHER AND HELP DECIDE WHERE DATA SHOULD GO NEXT.



PATHS ARE THE ROUTES THAT PACKETS TAKE ACROSS THE INTERNET. DIFFERENT PACKETS CAN TAKE DIFFERENT ROUTES.



REDUNDANCY MEANS THERE ARE MULTIPLE POSSIBLE PATHS FOR DATA, SO THE NETWORK HAS BACKUP OPTIONS.



FAULT TOLERANCE MEANS THE INTERNET CAN KEEP WORKING EVEN IF ONE PART FAILS, BECAUSE PACKETS CAN USE ANOTHER PATH.

- BANDWIDTH IS THE AMOUNT OF DATA THAT CAN BE SENT IN A CERTAIN AMOUNT OF TIME.
- LATENCY IS THE DELAY BEFORE DATA BEGINS TO ARRIVE.

HIGH BANDWIDTH HELPS WITH LARGE AMOUNTS OF DATA, WHILE LOW LATENCY IS IMPORTANT FOR THINGS LIKE GAMING AND VIDEO CALLS.

THESE TWO IDEAS ARE DIFFERENT. A CONNECTION CAN HAVE HIGH BANDWIDTH BUT STILL FEEL SLOW IF LATENCY IS HIGH. BOTH AFFECT HOW WELL THE INTERNET WORKS FOR DIFFERENT ACTIVITIES.

BANDWIDTH & LATENCY

PROTOCOLS

Protocols are rules that devices follow to communicate. They make sure data is sent and received in a standard way. The Internet depends on shared protocols so devices from different companies and systems can work together. Without protocols, devices would not know how to format, send, receive, or interpret data correctly. Shared rules make global communication possible.

1

IP (Internet Protocol)

Assigns unique addresses to devices and specifies where data should be delivered

2

TCP (Transmission Control Protocol)

helps make sure packets arrive correctly and in order.

3

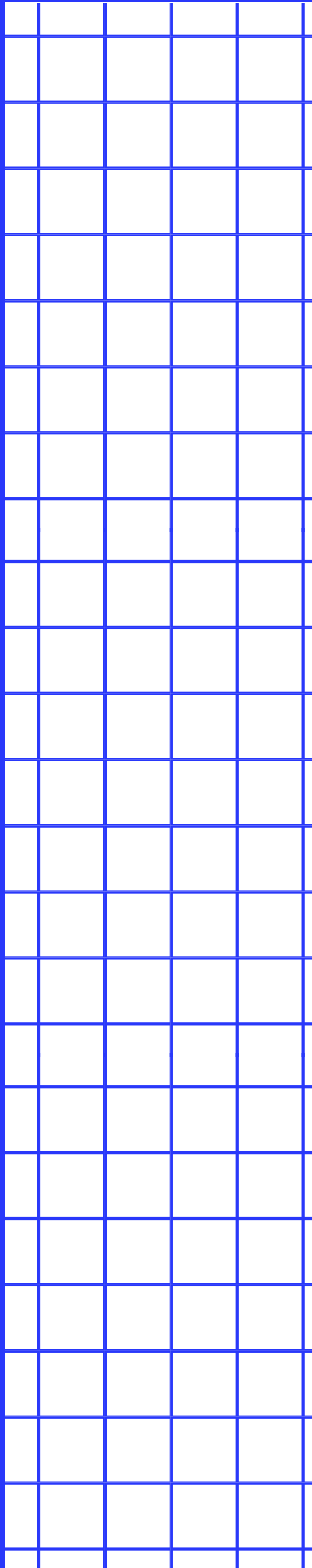
DNS (Domain Name System)

Translates website names into IP addresses so devices can find the correct server.

The Internet uses open standards, which are publicly available rules. These standards let different devices and systems communicate. Because of this design, the Internet is scalable, meaning it can keep growing to support more users, devices, and services.

No single company owns the whole Internet. Many organizations manage their own networks, and those networks connect through shared rules and standards.

OPEN STANDARDS AND SCALABILITY





REAL LIFE EXAMPLES

When you watch YouTube or Netflix, video data is broken into packets and sent across the Internet. Routers direct those packets through networks until they reach your device. Good bandwidth helps the video play smoothly. Low bandwidth can cause buffering or lower quality.

Apps like Zoom, FaceTime, and Google Meet send audio and video as packets in real time. Low latency is important because delays can interrupt conversation. Stable Internet connections improve call quality and make communication clearer.

Online games constantly send updates between player devices and game servers. This includes movement, actions, scores, and other live information. Low latency matters because even small delays can affect gameplay. Routers and packet switching help keep the game connected.

Map apps use the Internet to get directions, traffic updates, and estimated travel times. Your phone sends requests to servers, and the servers send data back. Live navigation depends on this constant exchange of information.

1. MULTIPLE CHOICE QUESTION

2 mins • 1 pt

Which of the following best explains how messages are typically transmitted over the Internet?

- ☐ The message is broken into packets that are transmitted in a specified order. Each packet must be received in the order it was sent for the message to be correctly reassembled by the recipient's device.
- ☐ The message is broken into packets. The packets can be received in any order and still be reassembled by the recipient's device.
- ☐ The message is broken into two packets. One packet contains the data to be transmitted and the other packet contains metadata for routing the data to the recipient's device.
- ☐ The message is transmitted as a single file and received in whole by the recipient's device.

2. MULTIPLE CHOICE QUESTION

2 mins • 1 pt

Which of the following is a primary reason for the use of open protocols on the Internet?

- ☐ Open protocols allow devices to specify how data packets are to be routed on the Internet in advance.
- ☐ Open protocols ensure that all data transmission on the Internet is kept secure.
- ☐ Open protocols ensure that all Internet users are provided connections with equal bandwidth.
- ☐ Open protocols provide a way to standardize data transmission between different devices.

1. Which of the following is true about packets?

- (A) Packets will travel along the fastest path from the sending to receiving computer.
- (B) Packets can arrive at the receiving computer in a different order than sent by the sending computer.
- (C) When sent by a sending computer packets are guaranteed to arrive at the receiving computer.
- (D) One sending file will contain one packet.

2. Which characteristics of internet protocols have helped fuel the rapid growth of the internet?

Select two answers.

- (A) Internet standards cannot be altered, so different companies can focus on development.
- (B) Internet protocols are open (nonproprietary).
- (C) Internet protocols are designed to be useful for a growing population of users.
- (D) The design of internet protocols involves large profits that motivate companies to innovate internally with no outside feedback.

6. Many types of computing devices made by many types of companies can all access the internet. Which of the following makes this access possible?

- (A) Redundancy
- (B) Open standards and protocols
- (C) Fault tolerance
- (D) Large bandwidth

<https://wayground.com/admin/quiz/623859ae06fb4b001e05e22c/ap-csp-internet-fault-tolerance-41-42>

<https://www.studocu.com/en-us/document/lambert-high-school/ap-precalculus/ap-csp-big-idea-4-prac/98021151>



**THANK YOU
FOR
LISTENING**